

Edge-Count Separation and Residual Orbit Collisions in a Nonlattice Binary Suspension

Route-A structural certificate C135

Abstract

We refine a symbol-count suspension clock to a directed-edge roof on the full binary shift. An exact four-variable matrix determinant gives the rooted trace law and primitive-orbit product at every period. The nonlattice roof $(1, \sqrt{2}, \sqrt{3}, \sqrt{6})$ separates every distinct directed edge-count sector, including the requested words 000111 and 001011. It remains noninjective on primitive orbits: 001011 and 001101 collide, and a flow-conservation theorem proves that all periodic data are blind to off-diagonal orientation. The construction is source-side and dynamical, not an arithmetic or target-facing factorization.

1 Formal determinant and all-period product

Let Σ be the two-sided full binary shift and assign the ordered edge roof

$$\tau = \begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{3} & \sqrt{6} \end{pmatrix}.$$

The suspension identifies $(x, t + \tau_{x_0 x_1})$ with $(\sigma x, t)$. For formal edge variables put

$$M(x) = \begin{pmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \end{pmatrix}, \quad \Delta(x) = \det(I - M(x)).$$

Direct calculation gives

$$\Delta(x) = 1 - x_{00} - x_{11} + x_{00}x_{11} - x_{01}x_{10}. \quad (1)$$

For every $n \geq 1$, matrix multiplication yields the exact rooted-word law

$$\text{Tr } M(x)^n = \sum_{\substack{w \text{ rooted closed} \\ |w|=n}} \prod_{i,j} x_{ij}^{N_{ij}(w)}. \quad (2)$$

Expanding $-\log \det(I - M) = \sum_{n \geq 1} \text{Tr}(M^n)/n$ and grouping each rooted word by its unique primitive root and repetition proves, in the total-edge-degree completion,

$$\Delta(x) = \prod_{[\gamma] \text{ mprimitive}} \left(1 - \prod_{i,j} x_{ij}^{N_{ij}(\gamma)} \right). \quad (3)$$

There are only finitely many primitive necklaces of any fixed total degree, so every coefficient of (3) receives finitely many contributions; the formal product is therefore well defined. Equations (2)–(3) hold at every period.

2 Nonlattice specialization

Substitute $x_{ij} = e^{-s\tau_{ij}}$. Then

$$d_\tau(s) = 1 - e^{-s} - e^{-\sqrt{6}s} + e^{-(1+\sqrt{6})s} - e^{-(\sqrt{2}+\sqrt{3})s}, \quad (4)$$

and (3) specializes to

$$d_\tau(s) = \prod_{[\gamma]} (1 - e^{-s\ell(\gamma)}), \quad \ell = N_{00} + \sqrt{2}N_{01} + \sqrt{3}N_{10} + \sqrt{6}N_{11}.$$

If $h > 0$ is the unique solution of $\rho((e^{-h\tau_{ij}})_{ij}) = 1$, the logarithmic trace and primitive expansions converge absolutely for $\text{Re } s > h$. Indeed, for $\sigma > h$ the positive matrix $M_\tau(\sigma) = (e^{-\sigma\tau_{ij}})$ has spectral radius below one, while absolute values at s are bounded entrywise by $M_\tau(\sigma)$. Hence $\sum_n \text{Tr}(M_\tau(\sigma)^n)/n$ converges and dominates both regroupings. The explicit finite exponential polynomial (4) is entire, so its reciprocal is meromorphic. The fixed cycles $[0]$ and $[1]$ have lengths 1 and $\sqrt{6}$; their irrational ratio rules out a common clock lattice and proves that the roof is nonlattice.

3 Exact edge-sector separation

The numbers $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ form the rational basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. For completeness, applying the four independent sign changes $\sqrt{2} \mapsto \pm\sqrt{2}$ and $\sqrt{3} \mapsto \pm\sqrt{3}$ to a rational linear relation and adding with signs isolates each coefficient. Equality of two roof lengths therefore forces equality of their entire directed-edge-count vectors. At period six,

word	$(N_{00}, N_{01}, N_{10}, N_{11})$	rooted multiplicity	length
000111	$(2, 1, 1, 2)$	6	$2 + \sqrt{2} + \sqrt{3} + 2\sqrt{6}$
001011	$(1, 2, 2, 1)$	12	$1 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$

Their difference is $1 - \sqrt{2} - \sqrt{3} + \sqrt{6} \neq 0$. Thus the two words, which have the same symbol population, occupy different edge sectors. Under C130's destination-symbol control $\tau_{ij} = \rho_j$ with $\rho = (1, \sqrt{2})$, both instead have length $3 + 3\sqrt{2}$.

The trace multiplicities also have a direct orbit explanation. A primitive period-six necklace contributes its six rooted rotations. There is one necklace in the $(2, 1, 1, 2)$ sector, giving coefficient 6, and two necklaces in the $(1, 2, 2, 1)$ sector, giving coefficient 12.

4 The residual collision is structural

The primitive words 001011 and 001101 are not cyclic rotations, but both have edge vector $(1, 2, 2, 1)$ and hence the same roof time. They are the first same-edge-count primitive collision, at period six.

This obstruction holds at every period. In any closed binary word, every crossing from 0 to 1 must be balanced by a crossing from 1 to 0; hence

$$N_{01} = N_{10}. \quad (5)$$

Every periodic length and determinant coefficient consequently depends on the off-diagonal roof only through $\tau_{01} + \tau_{10}$, never through $\tau_{01} - \tau_{10}$. Ordered formal variables preserve the accounting, but closed-orbit data themselves cannot recover orientation asymmetry.

Progress over C130. C130 separated symbol-population clock sectors but aggregated 000111 and 001011. C135 refines the same exact all-period determinant to complete directed-edge-count sensitivity and separates that pair. It deliberately retains collisions inside one edge sector and proves the exact observable quotient rather than claiming orbit injectivity.

5 Validation and Route-A boundary

The replay prefix contains 2,046 rooted words and 226 primitive cycles through period ten; the theorem has no cutoff. A standard-library checker independent of the producer passes 2,121 assertions, including (5) for every replay word. An independent SymPy reconstruction passes 37 checks; byte replay and all 43 hostile cases (42 repaired-hash plus one stale-hash case) pass.

The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL), overall ROUTE_A_EXPLORATORY. The product in (3) is a symbolic-dynamical primitive-orbit identity, not an arithmetic Euler product. We claim no target divisor or global analytic comparison, prime-like correspondence, arithmetic/local data, Euler factors, root numbers, automorphy, natural self-adjoint or unitary lift, Hilbert–Pólya operator, or Riemann-zero relation. Route B is unauthorized; the literal firewall is NO_BAD_EULER_OR_ROOT_NUMBER.